EXPERIMENT 7:

AIM:

To create a Pivot Table using Pandas and find the **maximum** and **minimum** sale value of each item.

INPUT:

Item,Salesman,Sale_Value

Pen,John,120

Pen,Alice,150

Pencil,John,80

Pencil,Alice,100

Eraser,John,50

Eraser,Alice,60

Notebook,John,200

Notebook,Alice,250

PROGRAM:

exp7.py > ...

```
import pandas as pd
df = pd.read_csv(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp7_data.csv")
pivot_table = pd.pivot_table(
    df,
    values="Sale_Value",
    index="Item",
    aggfunc=["max", "min"]
)
print("Maximum and Minimum Sale Value")
print(pivot_table)
```

OUTPUT:

```
Maximum and Minimum Sale Value
                max        min
         Sale_Value Sale_Value
Item
Eraser           60         50
Notebook        250        200
Pen             150        120
Pencil          100         80
```

RESULT:

The Pivot Table was successfully created and displayed the **maximum** and **minimum** sale value for each item.

EXPERIMENT 8:

AIM:

To create a Pivot Table using Pandas and find the **total units sold** for each item.

INPUT:

Item,Salesman,Units_Sold

Pen,John,10

Pen,Alice,15

Pencil,John,20

Pencil,Alice,25

Eraser,John,8

Eraser,Alice,12

Notebook,John,5

Notebook,Alice,7

PROGRAM:

```
exp8.py > ...
import pandas as pd
df = pd.read_csv(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp8_data.csv")
pivot_table = pd.pivot_table(
    df,
    values="Units_Sold",
    index="Item",
    aggfunc="sum"
)
print("Item-wise Units Sold")
print(pivot_table)
```

OUTPUT:

```
Item-wise Units Sold
          Units_Sold
Item
Eraser            20
Notebook          12
Pen               25
Pencil            45
```

RESULT:

The Pivot Table was successfully created and displayed the **item-wise total units sold** for each item.